

KUO, CHEN CHIH 郭晨志

Chemical & Materials R&D Engineer • Battery Specialist

b29374665@gmail.com

b29374665-design.github.io

Seeking EHS Engineer

Xinzhuang, New Taipei • Open to relocation / shifts

Profile

M.S. candidate in Chemical Engineering at NTUST, specializing in **electrochemistry, ultra-low-temperature electrolytes** and **lithium battery reliability / aging**; co-author on a **Nature Energy** submission.

Coursework in Nanomaterials and Applications (lithography, nanoimprint lithography, self-assembly); responsible for TGA thermal measurements in the Nature Energy work. Experienced in managing over NTD 2M of equipment procurement and the core thermal instruments (TGA Q500, DSC 2500) at the SEED Center, and independently responsible for an industry project on long-life Li-SOCl₂ cells for smart water meters (cell screening and EIS / aging analysis).

EHS Licenses

Coursework revealed a gap: understanding reaction yield and heat balance is not enough without knowing material logistics, on-site waste discharge and emergency response. I pursued the following EHS credentials part-time — evenings and weekends — during graduate study.		
✓	Dedicated Wastewater and Sewage Treatment Specialist (Class B)	Certified
✓	Indoor Air Quality Maintenance Management Specialist	Certified
✓	Waste Disposal Technician (Class A)	Certified
○	Occupational Safety and Health Management Personnel (Level B)	2026 / 11 • mid-Nov exam scheduled
○	Occupational Safety Management Specialist (Level A)	2027 / 03 • exam scheduled

Experience

2025 — Present	Sustainable Electrochemical Energy Development (SEED) Center, NTUST Lab Equipment Procurement & Asset Maintenance Manager
	• Procurement (>NTD 2M): Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes and low-temperature baths. • Instrument maintenance: Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500. • Maintained close, responsive contact with vendor engineers (Waters, TA Instruments, Uni-Onward), frequently securing on-site repair within 24 hours of a fault report — significantly reducing instrument downtime.
	Summer 2022 New Taipei City Dept. of Urban Development Summer Intern, Housing Development Section
2021 — 2024	• Reviewed rental subsidy applications and indexed confidential records in government databases; handled frontline public inquiries, building strong familiarity with administrative workflows and SOPs.
	Tunghai University Student Housing / GEPT Resident Assistant (2 yrs) / Exam Staff (part-time)
	• Oversaw safety and daily order in a 100-resident dormitory, mediating resident disputes and liaising with university administration; as part-time exam staff, handled test-paper check-in with zero errors.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker

Project lead

Screening methodology

Designed full-stage test protocols, experimental workflows and statistical validation models.

Database & SOP

Established the test database and standardized high-throughput screening procedures (SOP).

Advanced analysis

EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.

Training & tech transfer

Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Energy Journal submission — battery thermal stability study

Co-author Responsible for TGA thermal measurement

Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

Education

2025 – 2027 (expected)

National Taiwan University of Science and Technology

M.S. in Chemical Engineering

Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.

2020 – 2024

Tunghai University

B.S. in Chemical & Materials Engineering

Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

Skills & Instruments

Electrochemistry & batteries

Impedance spectroscopy (EIS)

Battery aging analysis

Reliability analysis

Low-temp electrolytes

Characterization

Thermal analysis (TGA / DSC)

Synchrotron (TLS20A / TPS27A / 45A)

Rheology (ARES-G3)

Viscometer & densimeter (DMA5002)

Process & operations

Lithography (nanoimprint / self-assembly)

Equipment procurement & maintenance

Software

 Aspen Plus V14

 Aspen Dynamics

 Microsoft Word

 Microsoft Excel

 Microsoft PowerPoint

 OriginPro

 TA Universal Analysis

 TA TRIOS

Conferences & Training

Presenter

2026 TWiChE Annual Meeting

Invited training

TA Waters User Seminar

Hands-on instrument operation